THE INDUCED LEAK

Electromagnetic Side-Channel Attack Through Inductive Coupling

Comprehensive Project Roadmap

ENSTA Algiers

Department of Computer Science Fundamentals (MI2)

Course Director: Dr. Cheggou Rabéa

Academic Year 2025–2026

March 31, 2026

Contents

1 Phase 1: Project Foundation & Contextualization **5**
1.1 Problem Statement . . . 5
1.1.1 The Security Threat . . . 5
1.2 Historical Precedents . . . 6
1.3 Why This Matters Now . . . 6

2 Phase 2: Mathematical & Physical Foundation **7**
2.1 Maxwell's Equations: Complete Foundation . . . 7
2.2 Faraday's Law: Attack Mechanism . . . 7
2.2.1 Magnetic Flux Definition . . . 7
2.2.2 Mutual Inductance . . . 7
2.2.3 Mutual Inductance Calculation for Parallel Cables . . . 8
2.3 Digital Binary Signal Model . . . 8
2.3.1 Induced Attack Signal . . . 8
2.3.2 Transition Detection Table . . . 8
2.3.3 Reconstruction Algorithm . . . 9
2.4 Signal Decay Model with Distance . . . 9
2.4.1 Magnetic Field Around Straight Cable . . . 9
2.4.2 Distance-Dependent EMF . . . 9
2.4.3 Signal-to-Noise Ratio (SNR) . . . 9
2.4.4 Electromagnetic Shielding Effect . . . 9
2.5 Frequency Domain Analysis (Fourier) . . . 9

3 Phase 3: Computer Science & Cybersecurity **10**
3.1 Complete Attack Architecture . . . 10
3.2 Signal Processing Pipeline . . . 11
3.3 Cybersecurity Classification . . . 11

4 Phase 4: State of the Art & Research Gaps **12**
4.1 Key Scientific References . . . 12
4.2 Unsolved Scientific Problems . . . 12

5 Phase 5: Original Innovations – Your Personal Contribution **13**
5.1 Innovation 1: Adaptive Threshold Algorithm . . . 13
5.2 Innovation 2: ICA Signal Separation . . . 13
5.3 Innovation 3: Protocol-Agnostic Decoder . . . 13
5.4 Innovation 4: Encrypted Data Leakage via Traffic Analysis . . . 14

6 Phase 6: Simulation & Implementation **15**
6.1 Proposed Code Structure (Python) . . . 15
6.2 Required Libraries . . . 15
6.3 Simulation Outputs . . . 16

7 Phase 7: Final Report **17**
7.1 Report Structure (40–60 Pages) . . . 17
7.2 Evaluation Criteria . . . 17

8 Project Timeline & Milestones **18**

9 Final Checklist Before Submission **18**

A Equation Summary Table **20**

Abstract

This document presents a comprehensive research roadmap for "The Induced Leak" project, an electromagnetic side-channel attack exploiting inductive coupling between parallel cables. The project bridges fundamental physics (Maxwell's Equations, Faraday's Law) with cybersecurity applications (side-channel attacks, signal processing). This roadmap covers theoretical foundations, mathematical derivations, state-of-the-art analysis, proposed innovations, simulation architecture, and implementation guidelines. The work aligns with ENSTA MI2 curriculum requirements while contributing original research to the physical layer security domain.

1 Phase 1: Project Foundation & Contextualization

Project Definition

Title	The Induced Leak: Electromagnetic Side-Channel Analysis of Parallel Cable Crosstalk
Domain	Physical Layer Security / Electromagnetic Compatibility / Side-Channel Attacks
Core Physics	Faraday's Law of Induction, Mutual Inductance, Maxwell's Equations
Core CS	Signal Processing, Binary Edge Detection, Cryptographic Key Extraction
Application	TEMPEST-style Attacks, Cable Shielding Design, Secure Infrastructure

1.1 Problem Statement

Central Research Question:
Can an external attacker, without physical contact, reconstruct digital data transmitted through Ethernet or USB cables using only Faraday's Law of electromagnetic induction?

1.1.1 The Security Threat

Sensitive data transmitted through copper cables generates electromagnetic fields. These fields can be captured **without physical contact** through inductive coupling, revealing:

- Encryption keys
- Passwords and credentials
- Classified communications
- Financial transactions

1.2 Historical Precedents

Table 1: Historical Electromagnetic Side-Channel Attacks

Incident	Year	Impact	Lesson Learned
TEMPEST Program (NSA)	1950s	Classified EM shielding standard	Threat exists for decades
Van Eck Phreaking	1985	Monitor emission interception	Screens broadcast data
USBee Attack	2017	USB data exfiltration via EM	USB cables act as antennas
LANtenna Attack	2019	Ethernet cable as transmitter	Wired networks not secure
Deep Learning EM Analysis	2023	AI-powered signal classification	Attacks becoming intelligent

1.3 Why This Matters Now

Current Relevance

- **IoT Devices:** Often lack electromagnetic shielding
- **High-Speed Data:** Stronger electromagnetic emissions
- **Quantum-Resistant Encryption:** Useless if keys leak physically
- **5G/6G Infrastructure:** Increased electromagnetic density

2 Phase 2: Mathematical & Physical Foundation

Core Physics Equations

All equations derived from Maxwell's Equations (Module 2, Course Material)

2.1 Maxwell's Equations: Complete Foundation

In free space ($\rho = 0$, $\vec{j} \neq 0$):

$$\nabla \cdot \vec{E} = 0 \quad \text{(Gauss's Law – No free charges)} \tag{1}$$

$$\nabla \cdot \vec{B} = 0 \quad \text{(Gauss's Law – No magnetic monopoles)} \tag{2}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{(Faraday's Law – Attack core)} \tag{3}$$

$$\nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{(Ampère-Maxwell Law)} \tag{4}$$

Critical Chain: Current in victim cable ($\vec{j}$) $\rightarrow$ Generates $\vec{B}$ (Ampère) $\rightarrow$ Varying $\vec{B}$ with data $\rightarrow$ Induces EMF in attacker cable (Faraday).

2.2 Faraday's Law: Attack Mechanism

$$\boxed{e = -\frac{d\Phi}{dt}}$$

Where:

- e = Induced Electromotive Force (EMF) in Volts
- Φ = Magnetic Flux through receiver loop (Weber)
- t = Time (seconds)

2.2.1 Magnetic Flux Definition

$$\Phi = \int_S \vec{B} \cdot d\vec{S} = B \cdot A \cdot \cos(\theta) \tag{5}$$

2.2.2 Mutual Inductance

$$e = -M \frac{dI}{dt} \tag{6}$$

Where M = Mutual Inductance between cables (Henry).

2.2.3 Mutual Inductance Calculation for Parallel Cables

$$M = \frac{\mu_0 \cdot l}{2\pi} \ln\left(\frac{d+a}{d}\right) \tag{7}$$

Where:

- l = Length of shared cable (meters)
- d = Distance between centers (meters)
- a = Radius of receiver loop (meters)

2.3 Digital Binary Signal Model

$\mathrm{I(t)} = \sum_n a_n \cdot p(t - nT_b)$

Where:

- $a_n \in \{0, 1\}$ = Binary data bits
- T_b = Bit period (seconds)
- $p(t)$ = Pulse shape function

2.3.1 Induced Attack Signal

$$e(t) = -M\frac{dI}{dt} = -M\sum_n a_n \cdot p'(t - nT_b) \tag{8}$$

2.3.2 Transition Detection Table

Table 2: Binary Transition Detection via Induced EMF

Transition	$\frac{dI}{dt}$	Induced EMF (e)	Detected Bit
$0 \to 0$	0	0	No change
$0 \to 1$	+Large	+Pulse	Rising Edge
$1 \to 0$	-Large	-Pulse	Falling Edge
$1 \to 1$	0	0	No change

2.3.3 Reconstruction Algorithm

```
if e > +threshold:
    bit = 1  # Rising Edge
elif e < -threshold:
    bit = 0  # Falling Edge
else:
    # No Transition
    pass
```

Listing 1: Edge Detection Logic

2.4 Signal Decay Model with Distance

2.4.1 Magnetic Field Around Straight Cable

$$B(r) = \frac{\mu_0 \cdot I}{2\pi r} \tag{9}$$

2.4.2 Distance-Dependent EMF

$$e(r) = -M(r) \cdot \frac{dI}{dt} \propto \frac{1}{r} \cdot \frac{dI}{dt} \tag{10}$$

2.4.3 Signal-to-Noise Ratio (SNR)

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{M^2 \cdot (dI/dt)^2}{N_0 \cdot B_w} \tag{11}$$

Where N_0 = Noise spectral density, B_w = Filter bandwidth.

2.4.4 Electromagnetic Shielding Effect

$$e_{\text{shielded}} = e_{\text{unshielded}} \cdot 10^{-S/20} \tag{12}$$

Where S = Shielding effectiveness (dB).

2.5 Frequency Domain Analysis (Fourier)

E(f) = F{e(t)} = -j $\cdot 2\pi f \cdot M \cdot I(f)$

Critical Insight: Higher frequency (faster data rates) = Stronger induced signal. This explains why 10 Gbps Ethernet cables are more vulnerable than slower cables.

3 Phase 3: Computer Science & Cybersecurity

3.1 Complete Attack Architecture

Table 3: Attack Layer Architecture

Layer	Component	Technology
Physical Layer	Magnetic reception loop	Antenna design
Acquisition	Analog-to-Digital Converter (ADC)	Sampling rate: $10\times T_b$
Pre-processing	Bandpass filter	Noise removal
Amplification	Low-Noise Amplifier (LNA)	Boost weak signal
Edge Detection	Derivative algorithm	Identify $0\to1$ and $1\to0$
Reconstruction	Finite state machine	Restore bit sequence
Analysis	Protocol parser	USB / Ethernet / HDMI
Data Extraction	Cybersecurity algorithm	Keys, passwords

3.2 Signal Processing Pipeline

Table 4: Signal Processing Stages

Stage	Algorithm	Purpose
1. Acquisition	ADC @ 10× bit rate	Capture analog induced signal
2. Filtering	Bandpass ($f_{data} \pm 20\%$)	Remove out-of-band noise
3. Normalization	Z-Score Normalization	Reduce variance impact
4. Edge Detection	Derivative + Adaptive Threshold	Identify transitions
5. Reconstruction	Finite state machine	Rebuild bit sequence
6. Verification	BER calculation	Measure attack accuracy
7. Decoding	Protocol parser	Extract meaningful data

3.3 Cybersecurity Classification

Security Metrics

Attack Type	Side-Channel Attack (Physical Layer)
Access Required	Proximity (no physical contact)
Stealth	Completely passive – Cannot be traditionally detected
Detection	Very difficult (no network modification)
CWE Classification	CWE-203: Observable Discrepancy
CVSS Score Estimate	6.8 (Medium-High)
Mitigation	EM shielding, Filtering, Constant-time encoding

4 Phase 4: State of the Art & Research Gaps

4.1 Key Scientific References

Table 5: Major Research Works in EM Side-Channel Attacks

Research	Year	Achievement	Limitation
TEMPEST Standards (NSA)	1950s–Now	Military EM shielding protocols	Classified, expensive
Vuagnoux & Pasini	2009	Wired keyboard keystroke capture	Distance $<$ 20cm
USBee Attack	2017	USB data exfiltration via EM	Range $<$ 1m
LANtenna Attack	2019	Ethernet cable as antenna	Requires malware on target
DeepEM (IEEE)	2023	Deep Learning signal classification	High computational cost

4.2 Unsolved Scientific Problems

Open Research Challenges

Problem	Why Unsolved	Difficulty
Long-range detection	Signal decays as $1/r$	
Multi-cable separation	Emission entanglement between cables	
Encrypted traffic analysis	Cannot decrypt directly	
Real-time detection	Processing latency exceeds transmission time	
Unified countermeasure verification	No standard testing protocol	

5 Phase 5: Original Innovations – Your Personal Contribution

5.1 Innovation 1: Adaptive Threshold Algorithm

Instead of fixed threshold for edge detection, use adaptive statistical model:

$$\tau(t) = \mu_{\text{local}}(t) + k \cdot \sigma_{\text{local}}(t)$$

Where:

- μ_{local} = Local time window signal mean
- σ_{local} = Local standard deviation
- k = Adaptation coefficient (tuned via machine learning)

Advantage: Adapts to environmental noise changes without manual recalibration.

5.2 Innovation 2: ICA Signal Separation

For multiple cables, use Independent Component Analysis:

$$\vec{x}(t) = A \cdot \vec{s}(t) \tag{13}$$

Where:

- $\vec{x}(t)$ = Mixed sensor signals
- A = Unknown mixing matrix
- $\vec{s}(t)$ = Original source signals

ICA Goal: Recover $\vec{s}(t)$ without knowing A in advance – mathematically separate each cable.

5.3 Innovation 3: Protocol-Agnostic Decoder

Build machine learning model that recognizes **edge timing patterns** rather than protocol specifics, enabling USB, Ethernet, and HDMI decoding with single model.

5.4 Innovation 4: Encrypted Data Leakage via Traffic Analysis

Novel Contribution: Even encrypted data leaks information through:

- Transition timing patterns
- Edge frequency distribution
- Packet length distribution

This is a **Metadata Attack** – active research domain.

6 Phase 6: Simulation & Implementation

6.1 Proposed Code Structure (Python)

```
# faraday_attack_simulation/
#             core/
#                   signal_generator.py       # Digital signal generation
#                   em_model.py               # Inductive coupling model
#                   noise_model.py            # Noise model
#                   channel_model.py          # Distance and shielding effects
#             processing/
#                   filter.py                 # Filtering
#                   edge_detector.py          # Edge detection
#                   reconstructor.py          # Reconstruction
#                   ber_analyzer.py           # Error rate analysis
#             innovations/
#                   adaptive_threshold.py     # Innovation 1
#                   ica_separator.py          # Innovation 2
#                   traffic_analyzer.py       # Innovation 4
#             visualization/
#                 plots.py                  # Graphs and results
```

Listing 2: Project Directory Structure

6.2 Required Libraries

Table 6: Python Libraries and Roles

Library	Role
`numpy`	Numerical computation, signal arrays
`scipy.signal`	Filtering, edges, frequency transform
`matplotlib / seaborn`	Signal and result visualization
`scikit-learn`	Adaptive threshold, ICA
`pandas`	Experiment data management

6.3 Simulation Outputs

Table 7: Required Simulation Deliverables

Output	Description
Plot 1	Original binary signal vs. time
Plot 2	Induced EMF signal (with and without noise)
Plot 3	Filtered signal with detected edges
Plot 4	Reconstructed signal vs. original + BER
Plot 5	Success rate as function of distance
Plot 6	Success rate as function of SNR
Plot 7	Innovation comparison (fixed vs. adaptive algorithm)
Table 1	BER for different configurations
Table 2	Shielded vs. unshielded comparison

7 Phase 7: Final Report

7.1 Report Structure (40–60 Pages)

Table 8: Final Report Sections

Section	Weight	Content
1. Introduction	10%	Problem statement, motivation, objectives
2. Theoretical Framework	25%	Maxwell, Faraday, Mutual Inductance derivations
3. State of the Art	15%	Literature review, existing attacks, limitations
4. Methodology	20%	Your approach, innovations, algorithms
5. Implementation	15%	Code architecture, simulation details
6. Results	10%	Graphs, BER analysis, success rates
7. Countermeasures	5%	Technical recommendations for hardening
8. Conclusion	5%	Summary, future work, contributions

7.2 Evaluation Criteria

What Professors Look For

- **Physics Depth:** Correct use of Maxwell/Faraday equations, proper derivations
- **Mathematical Rigor:** Clear variable definitions, unit consistency, error analysis
- **CS Integration:** Signal processing algorithms, efficient code, proper architecture
- **Security Analysis:** Understanding of side-channel attack classification, CVSS
- **Innovation:** Your original contributions (not just textbook material)
- **Validation:** Simulation results match theoretical predictions
- **Professionalism:** Report formatting, citations, code documentation

8 Project Timeline & Milestones

Table 9: 12-Week Project Schedule

Week	Milestone	Deliverable
1–2	Literature Review	State of Art Document (10 pages)
3–4	Physics Modeling	Faraday/Mutual Inductance Equations Document
5–6	Simulation Core	Working Python code (Layers 1–3)
7–8	Attack Pipeline	Edge Detection + Decoding (Layers 4–5)
9	Results Generation	All plots and tables
10	Countermeasures	Technical Recommendations Document
11	Final Report	Complete PDF (40–60 pages)
12	Presentation	Slide Deck + Live Demo

9 Final Checklist Before Submission

Pre-Submission Verification

- ☐ All physics equations derived from Maxwell's fundamentals
- ☐ All variables defined with units
- ☐ Simulation code documented and reproducible
- ☐ Results include error analysis (BER, SNR)
- ☐ State of art includes 10+ academic references
- ☐ Countermeasures are technically specific (not generic)
- ☐ Original innovations clearly identified
- ☐ Report follows ENSTA formatting guidelines
- ☐ Presentation includes live simulation demo

Conclusion

Key Differentiator: The strongest aspect of your project is the **bridge between physics and cybersecurity**. Ensure every step in the attack chain is explicitly linked to a physical equation. This makes it genuine scientific research, not just a software project.

This roadmap provides complete coverage of all requirements from ENSTA MI2 Physics/Computer Vision course materials while enabling original research contributions to electromagnetic side-channel attack domain.

References

[1] Maxwell, J. C. (1865). "A Dynamical Theory of the Electromagnetic Field." *Philosophical Transactions of the Royal Society.*

[2] NSA. (1950s–Present). "TEMPEST Standards for Electromagnetic Shielding." *Classified Documents.*

[3] Van Eck, W. (1985). "Electromagnetic Radiation from Video Display Units: An Eavesdropping Risk?" *Computers & Security*, 4(4), 269-286.

[4] Guri, M., et al. (2017). "USBee: Smuggling Data from Air-Gapped Systems over USB Buses." *arXiv preprint arXiv:1706.05068.*

[5] Guri, M., et al. (2019). "LANtenna: Covert Channel over LAN Cables." *IEEE Access.*

[6] Various Authors. (2023). "Deep Learning for Electromagnetic Side-Channel Analysis." *IEEE Transactions on Information Forensics and Security.*

[7] Cheggou, R. (2025). "From Geometrical Optics to Wave Optics: Maxwell's Theory." *ENSTA Algiers MI2 Course Materials.*

[8] Cheggou, R. (2025). "Maxwell's Eureka: From Electricity to Light." *ENSTA Algiers MI2 Course Materials.*

[9] ENSTA Algiers. (2025). "Physics/Computer Vision Module: Workshop Exercises." *DFI-MI2.*

A Equation Summary Table

Table 10: Complete Equation Reference

Equation #	Formula	Module
1–4	Maxwell's Equations	2.1
2.2	Faraday's Law	2.2
5	Magnetic Flux	2.2
6	Mutual Inductance	2.2
7	Mutual Inductance Calculation	2.2
2.3	Binary Signal Model	2.3
8	Induced Attack Signal	2.3
9	Magnetic Field Decay	2.4
10	Distance-Dependent EMF	2.4
11	Signal-to-Noise Ratio	2.4
12	Shielding Effect	2.4
2.5	Fourier Transform	2.5
5.1	Adaptive Threshold	5.1
13	ICA Signal Separation	5.2